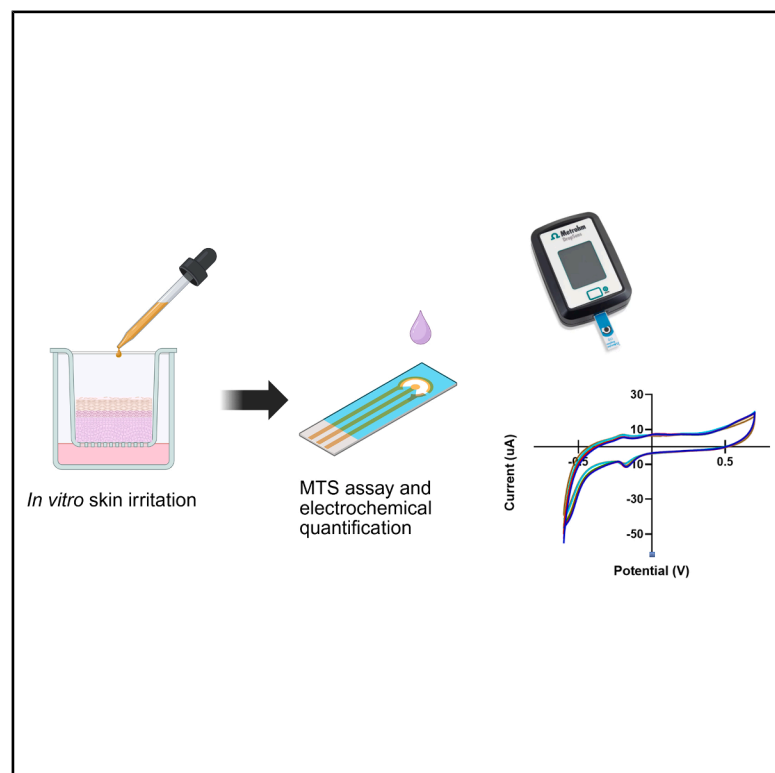


Electrochemical sensing of cell viability: A scalable platform from 2D monolayers to 3D tissue models

Graphical abstract



Authors

Ignacio Risueño, Sandra Navarro, Maarten van Os, Pablo Acedo, Leticia Valencia, Diego Velasco

Correspondence

Ivalenci@ing.uc3m.es (L.V.),
divelasc@ing.uc3m.es (D.V.)

In brief

Biological sciences; Bioengineering; Cell biology

Highlights

- An electrochemical platform was built to assess cell viability
- The sensor eliminates chemical and optical interferences
- Metabolic MTS reduction was validated in 2D and 3D skin models
- The work aligns with OECD skin toxicity regulatory standards



Article

Electrochemical sensing of cell viability: A scalable platform from 2D monolayers to 3D tissue models

Ignacio Risueño,^{1,7} Sandra Navarro,^{1,7} Maarten van Os,¹ Pablo Acedo,^{2,4,5,6} Leticia Valencia,^{1,3,*} and Diego Velasco^{1,3,8,*}¹Bioengineering Department, Universidad Carlos III de Madrid (UC3M), 28918 Leganés, Spain²Electronics Technology Department, Universidad Carlos III de Madrid (UC3M), 28918 Leganés, Spain³Fundación Instituto de Investigación Sanitaria de La Fundación Jiménez Díaz (IIS-FJD), 28040 Madrid, Spain⁴Departamento de Física, University of the Basque Country (UPV/EHU), 48080 Bilbao, Spain⁵EHU Quantum Center, University of the Basque Country (UPV/EHU), 48013 Bilbao, Spain⁶IKERBASQUE, Basque Foundation for Science, 48009 Bilbao, Spain⁷These authors contributed equally⁸Lead contact

*Correspondence: lvalenci@ing.uc3m.es (L.V.), divelasc@ing.uc3m.es (D.V.)

<https://doi.org/10.1016/j.isci.2026.117118>

SUMMARY

The trend toward animal-free testing in toxicology and drug development has accelerated the adoption of complex 3D *in vitro* models. However, standard colorimetric viability assays suffer from severe limitations due to chemical interferences. To address these challenges, we designed a versatile sensor platform based on screen-printed carbon electrodes to overcome optical bottlenecks by quantifying cell viability through the electrochemical reduction of MTS-formazan. The electrochemical sensor was validated in 2D human dermal fibroblasts and keratinocytes at different cell confluences. Importantly, it successfully assessed viability in both 2D cell monolayer cultures and reconstructed human epidermis exposed to an OECD-panel of irritant and non-irritant chemicals. The electrochemical approach provides a technologically robust and interference-free tool that enhances reliability in viability assays, overcoming the traditional limitations of colorimetric measurements. This methodology represents a transversal tool applicable to any tissue offering a robust and promising solution for high-throughput screening in pharmacology, cosmetics, and biomedical research.

INTRODUCTION

The path that a new drug candidate must follow to reach the market is a complex, multi-stage process that starts with the initial drug discovery, is followed by preclinical testing and finishes with an extensive clinical trial. Traditionally, the preclinical phase has served as the critical phase to test safety and efficacy, relying heavily on animal models to predict human response before going to the clinical stages.^{1–3} However, this traditional pipeline is facing significant ethical and economic transformation due to lower efficiency rates—where drugs successful in animals fail in humans—and growing regulatory pressure. A global paradigm shift is also driven by strict international frameworks restricting animal use. Landmark initiatives such as the EU Cosmetics Regulation (EC 1223/2009), the REACH Regulation (EC 1907/2006) for chemical safety, and the FDA Modernization Act 2.0 in the United States^{4,5} have set a worldwide precedent to minimize animal testing. Consequently, there is an urgent demand across the scientific community, as well as with pharmaceutical, chemical, and cosmetic industries for reliable, ethically approved *in vitro* methodologies capable of replacing animal models while ensuring robust safety standards.

To meet these demands, the field has moved beyond simplistic 2D cell monolayers, which often lack the physiological complexity required for accurate toxicity prediction. The industry standard is shifting toward 3D *in vitro* models, such as organoids, spheroids, and reconstructed tissues (cornea, skin, oral, airway, alveolar, kidney, liver, or intestine).^{6,7} These models have become indispensable tools, some of them even being validated by the OECD for regulatory decision-making.⁸ More specifically, reconstructed human epidermis (RHE) models, which are composed of differentiated keratinocyte layers, are relevant *in vitro* systems able to mimic the stratum corneum and barrier functions of real skin.⁹ These systems are valuable *in vitro* tools that offer ethical and physiologically relevant alternatives to animal models for chemical testing, drug permeation, and disease modeling. However, while the biological models have evolved significantly to mimic human physiology, the analytical techniques used to monitor them have lagged behind.

Currently, the assessment of cell viability in these tissues relies predominantly on colorimetric assays that measure metabolic activity, such as the reduction of the 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) to formazan salt. Mitochondrial NADH is the primary electron donor in enzymatic



reactions that reduce the yellow MTT tetrazolium salt into purple formazan crystals.¹⁰ Then, the amount of reduced product is measured using a spectrophotometer. MTT provides a reliable and standardized method, as it is internationally accepted for assessing cell viability, as well as a chemical classification method for irritation and corrosion exposure. However, it also presents several limitations when used in toxicity studies, such as possible interference with some chemicals leading to false positives, non-specific absorption, incomplete penetration in 3D models, or low chemical solubility.^{11–16} Furthermore, MTT can exhibit cytotoxicity due to the formation of insoluble formazan crystals within the cells, potentially altering cellular metabolism and leading to inaccurate viability measurements.¹¹

To overcome some of these limitations, the MTS assay was developed as a second-generation tetrazolium assay. The electrochemical quantification of cell viability is based on the reduction of the tetrazolium salt MTS [3-(4,5-dimethylthiazol-2-yl)-5-(3-carboxymethoxyphenyl)-2-(4-sulfophenyl)-2H-tetrazolium] into its soluble formazan derivative (Figure S1). This process occurs via a two-electron reduction mechanism.^{17,18} Unlike MTT, the MTS reagent directly produces a soluble formazan product, eliminating the need for solubilization steps and thereby reducing overall assay time, complexity, and accuracy. MTS has been used in several *in vitro* applications and for a wide range of cell types and organs. For instance, it has been effectively used in assessing the survival fraction of irradiated colon cancer cells,¹⁷ determining the cytotoxicity of contact lens care products on human corneal epithelial cells,^{19,20} and evaluating the effects of various compounds or devices on human dermal fibroblasts, keratinocytes,²¹ and RHE.^{22,23} However, regardless of these results, the deficiencies of the colorimetric technique commonly used in MTS assays for cell viability has led to the OECD's reluctance to accept this method due to susceptibility to interference.

Beyond conventional spectrophotometric techniques, electrochemical biosensors have emerged as powerful tools, for instance, in a wide range of tissue engineering applications due to their high sensitivity, rapid response, miniaturization potential, and direct electrical readout.²⁴ These platforms offer significant advantages (Table S1) for detecting a wide range of biomarkers and physiological changes.²⁵ For instance, electrochemical sensors have been developed for the non-invasive detection of skin analytes such as glucose,²⁶ lactate,²⁷ and pH²⁸ for continuous health monitoring or diagnostics. In the context of cytotoxicity and *in vitro* tissue engineering, electrochemical impedance spectroscopy (EIS) has been extensively used to monitor cell adhesion, proliferation, and barrier integrity in real-time, providing valuable insights into cellular health and tissue development.^{29–31} Furthermore, enzyme-based electrochemical biosensors have been applied to detect specific enzymes or metabolites indicative of cellular stress or damage in cell cultures.^{32,33} While these studies highlight the versatility of electrochemical sensing in cell-based assays and tissue engineering research, they still lack robust, interference-free, and high-throughput performance, especially for standardized cell viability endpoint studies in complex 3D human tissue models.

To address these limitations and exploit the inherent advantages of electrochemical detection, we developed a groundbreaking electrochemical sensor platform for the direct and

accurate measurement of cell viability, taking advantage of the inherent capabilities of electrochemical detection. This sensor aimed to measure the reduction of MTS to formazan more accurately than spectrophotometry techniques. Nicotinamide adenine dinucleotide (NADH), a physiologically relevant reducing agent,¹⁸ was used to validate the sensor, and further biological analyses were performed inhibiting its action with NADH-oxidase, in order to validate the reducing power to formazan.³⁴ The electrochemical sensor has demonstrated its ability to quantify cell confluence and cell viability in human skin fibroblasts and keratinocytes. The proposed method is described as a “scalable platform” due to its inherent compatibility with high-throughput architectures. This scalability ensures that the transition from bench-top proof-of-concept to automated industrial toxicity screening is technically straightforward, maintaining analytical performance while significantly increasing sample throughput. To demonstrate the broad applicability of this technology, we validated the sensor using RHE as a proof-of-concept model. We compared our electrochemical sensor against standard spectrophotometry under OECD TG 439 irritation protocols using a panel of reference chemicals. This approach effectively decouples the viability signal from optical constraints: it translates the metabolic reduction of MTS into a precise electric current rather than a color change. This method eliminates the need for crystal solubilization, avoids optical interference from the tissue or medium, and offers a straightforward path toward automation and high-throughput screening. While spectrophotometric MTS assays are prone to inaccuracies due to light scattering by tissue debris and optical interference from colored or turbid chemicals,³⁵ the proposed electrochemical detection cuts out these issues by taking advantage of the specific redox window of formazan, which remains independent of the sample's optical properties. The successful validation in this complex skin model serves as a robust demonstration of its potential as reproducible, non-destructive, and quantitative high-throughput system that can be applicable to any tissue model, enabling more accurate, automated, and animal-free drug development pipelines.

RESULTS

To address the limitations of conventional colorimetric assays for cell viability assessment, an alternative electrochemical method utilizing screen-printed carbon electrodes (SPCEs) was developed for the measurement of MTS reduction. The reduction of the MTS tetrazolium salt into its formazan derivative involves a two-electron/two-proton transfer that results in the opening of the tetrazole ring. This chemical transformation, thoroughly described in previous studies^{18,36} is the basis for both the optical change and the electrochemical signal reported here. The feasibility of this approach primarily depends on two critical aspects: first, the absence of interfering electrochemical signals from the cell culture media components, and second, the electrochemical detectability of the reduced MTS molecule signal.

Electrochemical characterization of MTS reduction and culture media interference

This initial section focuses on the study of the electrochemical background of commonly used biological media. For that

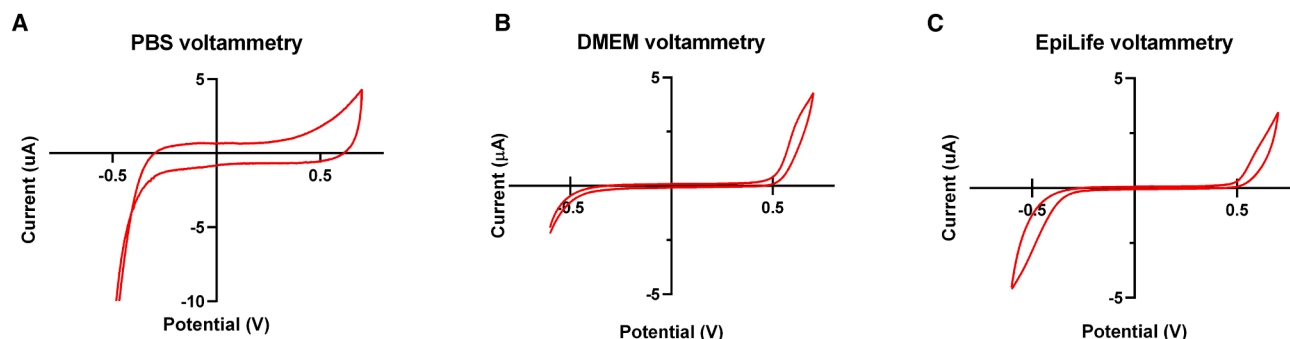


Figure 1. Cyclic voltammetry scans of various cell culture media components using SPCEs in a three-electrode configuration

(A) Phosphate buffered saline (PBS), (B) DMEM medium, and (C) EpiLife medium. All scans were performed at a scan rate of 50 mV/s. The absence of distinct oxidation or reduction peaks across the potential range (−0.6 to +0.7 V) confirms the electrochemical inertness of these media, ensuring no interference with subsequent MTS signal detection.

purpose, cyclic voltammetry (CV) was employed to characterize the electrochemical response of the individual components of the cell culture environment, namely, DMEM and EpiLife medium,³⁷ that is used for culturing N/TERT-1 and hDFs, respectively. As shown in Figure 1, no significant oxidation or reduction peaks were observed across the scanned potential range of −0.6 V to +0.7 V for any of the tested media. The absence of characteristic current responses within this electrochemical window confirms that these media components do not exhibit intrinsic electroactivity under the experimental conditions, thereby ensuring minimal interference with the subsequent detection of the MTS reduction product. This initial experiment is crucial for validating the specificity of the electrochemical sensor for MTS.

After the confirmation of electrochemical inactivity of the culture media, the electrochemical reduction of MTS mediated by NADH³⁸ was investigated using CV (Figure 2). NADH¹⁸ was employed as a model reducing agent to simulate the cellular metabolic activity that would typically convert MTS to its formazan product. The sensing principle relies on the mechanism where

metabolically active cells produce NADH, which reduces the MTS inner salt into a soluble formazan product. This formazan is the primary electroactive analyte in our system. During the CV scan, the formazan undergoes a 2-electron reduction at the working electrode, generating a cathodic peak between −0.2 and −0.1 V. Therefore, the peak current is directly proportional to the formazan concentration, which in turn reflects the amount of NADH produced by the cells.³⁹

As shown in Figure 2A, the CV obtained from MTS solutions reduced by varying concentrations of NADH revealed a characteristic negative current peak in the cathodic scan, specifically within the potential range of −0.3 to −0.1 V (highlighted by the red square). This peak corresponds to the electrochemical oxidation of the NADH-reduced formazan product of MTS. A more detailed examination of this region (Figure 2B) demonstrates a clear correlation between the NADH concentration and the magnitude of this reduction peak. Specifically, as the concentration of NADH increased from 0.01 to 0.1 mg/mL, a corresponding increase in the absolute magnitude of the current

MTS voltammetry with different [NADH]

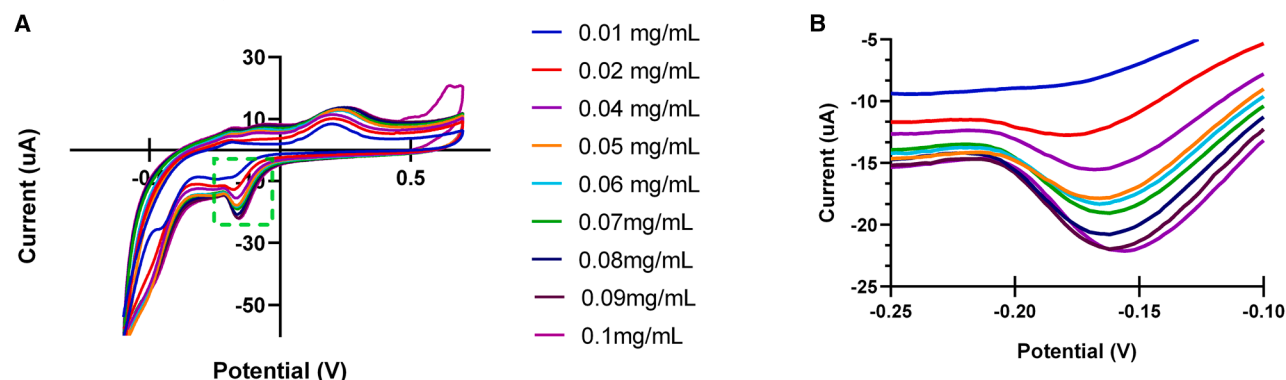


Figure 2. Cyclic voltammetry of MTS samples reduced with different NADH concentrations, performed using SPCEs at a scan rate of 50 mV/s

(A) The complete cyclic voltammograms demonstrating the presence of a current peak in the cathodic scan, located between −0.3 and −0.1 V (indicated by the red square), corresponding to the electrochemical signal of reduced MTS (formazan).

(B) A magnified view of the potential region where the MTS signal is detected. This zoomed-in section illustrates the concentration-dependent increase in the absolute current magnitude as NADH concentration, and consequently, the amount of reduced MTS increases. Plotted data represent the mean value of each repetition ($n = 7$).

peak was observed, indicating a stronger electrochemical signal with higher concentrations of reduced MTS.

The linear relationship between NADH concentration and MTS reduction was quantitatively obtained using both the conventional spectrophotometric method and the proposed electrochemical sensor, in the presence of EpiLife and DMEM culture media. As demonstrated in Figure 3, increasing concentrations of NADH (0.005–0.1 mg/mL) led to a proportional increase in MTS reduction.

For the conventional spectrophotometric method, a strong linear regression was observed between NADH concentration and mean absorbance for both EpiLife (Figure 3A, $R^2 = 0.9522$) and DMEM (Figure 3C, $R^2 = 0.9486$). Similarly, the electrochemical sensor exhibited excellent linearity, with the current intensity peak showing a strong linear correlation with NADH concentration in EpiLife (Figure 3B, $R^2 = 0.9493$) and DMEM (Figure 3D, $R^2 = 0.9667$).

Moreover, the sensor displayed a linear response with a calculated limit of detection (LOD) calculated based on the linear regression data from our CV measurements⁴⁰ and yielding a value of 0.0047 mg/mL. By using the standard deviation of the residuals ($Sy.x = 0.2292$) and the slope from this linear fit, we determined an LOD of practical sensitivity of our CV-based method. This level of sensitivity (Figure S3) demonstrates its suitability for quantitative cell viability monitoring in both 2D monolayers and complex 3D tissue models, where formazan production typically exceeds this threshold. The analytical performance of the sensor was further validated using differential pulse voltammetry (DPV), showing high sensitivity even at low NADH concentrations (Figure S4) and supporting the analytical performance of the proposed sensor.⁴¹

Furthermore, a critical aspect for validation of the proposed electrochemical method was to establish its correlation with the established spectrophotometric assay. Direct comparison of the data obtained from both methods revealed a strong linear correlation between the electrochemical current intensity and the spectrophotometric absorbance values. This was clearly shown in both EpiLife (Figure 3E, $R^2 = 0.9327$) and DMEM (Figure 3F, $R^2 = 0.9023$). The high R^2 values demonstrate that the electrochemical sensor provides results that are highly consistent with the gold standard colorimetric assay, reinforcing its potential as a reliable alternative for MTS reduction measurements.

To ensure that the measured current peak originates exclusively from the MTS-reduction product (formazan), several control experiments were conducted. As shown in Figure S2, no electrochemical activity was observed for the culture medium alone or for the individual components (MTS or NADH) within the scanned potential range. The characteristic reduction peak only emerged upon the combination of all reagents, confirming the selectivity of the platform for formazan quantification.

To evaluate the precision and reproducibility of the developed electrochemical sensor compared to the conventional spectrophotometric method, the percentage error in the measurements of MTS reduction was quantified. As shown in Figure 4, the electrochemical readings consistently exhibited improved precision. Statistical analysis revealed significant differences in the variability of results between the two methods ($p < 0.01$).

Specifically, for measurements conducted in EpiLife medium, the electrochemical sensor yielded a mean percentage error of

$1.73 \pm 1.19\%$ for the current intensity readings. In contrast, the spectrophotometric measurements in EpiLife showed a higher mean percentage error of $4.62 \pm 3.09\%$. Similarly, when evaluated in DMEM, the electrochemical method demonstrated a mean error of $2.63 \pm 1.64\%$, which was remarkably lower than the $5.25 \pm 3.44\%$ observed for the absorbance readings. The consistently smaller standard deviations and less dispersed data points associated with the electrochemical measurements (Figure 4) underscore higher consistency and precision.

Cell confluency and MTS electrochemical signal correlation

Following the initial validation of the electrochemical sensor and its correlation with the spectrophotometric method in cell-free systems, the applicability of the sensor to measure cell metabolic activity was investigated using varying cell confluency. This experiment aimed to confirm that the presence of cells, and their metabolic state, would not introduce interfering signals and that the established correlation between the two detection methods would be maintained in a biological context.

hDFs were seeded at different confluency percentages (10%–100%), and MTS reduction was subsequently measured using both the conventional spectrophotometric assay and the proposed electrochemical sensor. As presented in Figure 5, a strong linear correlation was observed between cell confluency and MTS reduction for both measurement techniques. The mean absorbance (OD) measured by spectrophotometry showed a good linear relationship with cell confluency (Figure 5A, $R^2 = 0.9684$). Similarly, the electrochemical current intensity peak also demonstrated linearity with cell confluency (Figure 5B, $R^2 = 0.9358$). These results confirm the ability of the sensor to quantitatively detect cellular metabolic activity and directly correlate it with cell density.

Moreover, the strong correlation between the two measurement methods was maintained in the presence of living cells. A robust linear relationship was observed when comparing the electrochemical current intensity readings directly against the spectrophotometric absorbance values (Figure 5C, $R^2 = 0.9056$). This high coefficient of determination indicates that the electrochemical sensor provides an equivalent and reliable readout of MTS reduction in a cellular environment.

To further demonstrate the causal link between cellular metabolism and the electrochemical response, a validation assay using a NADH production inhibitor was performed. As shown in Figures 5C and 5D, cells treated with NADH-oxidase exhibited a significant decrease in the reduction peak current compared to the positive control. To identify any potential artifacts caused by the solvent, a control with DMSO was included at the same concentration used in the inhibitor treatment, showing no significant impact on the signal. These results confirm that the electrochemical readout is directly dependent on the enzymatic reduction of MTS via cellular NADH-oxidase, reinforcing the mechanistic validity of our platform as previously suggested by similar mitochondrial inhibition studies.³⁴

Cell monolayer viability after chemical exposure

A crucial application for the proposed biosensor in skin irritation assessment is its ability to accurately determine cell viability

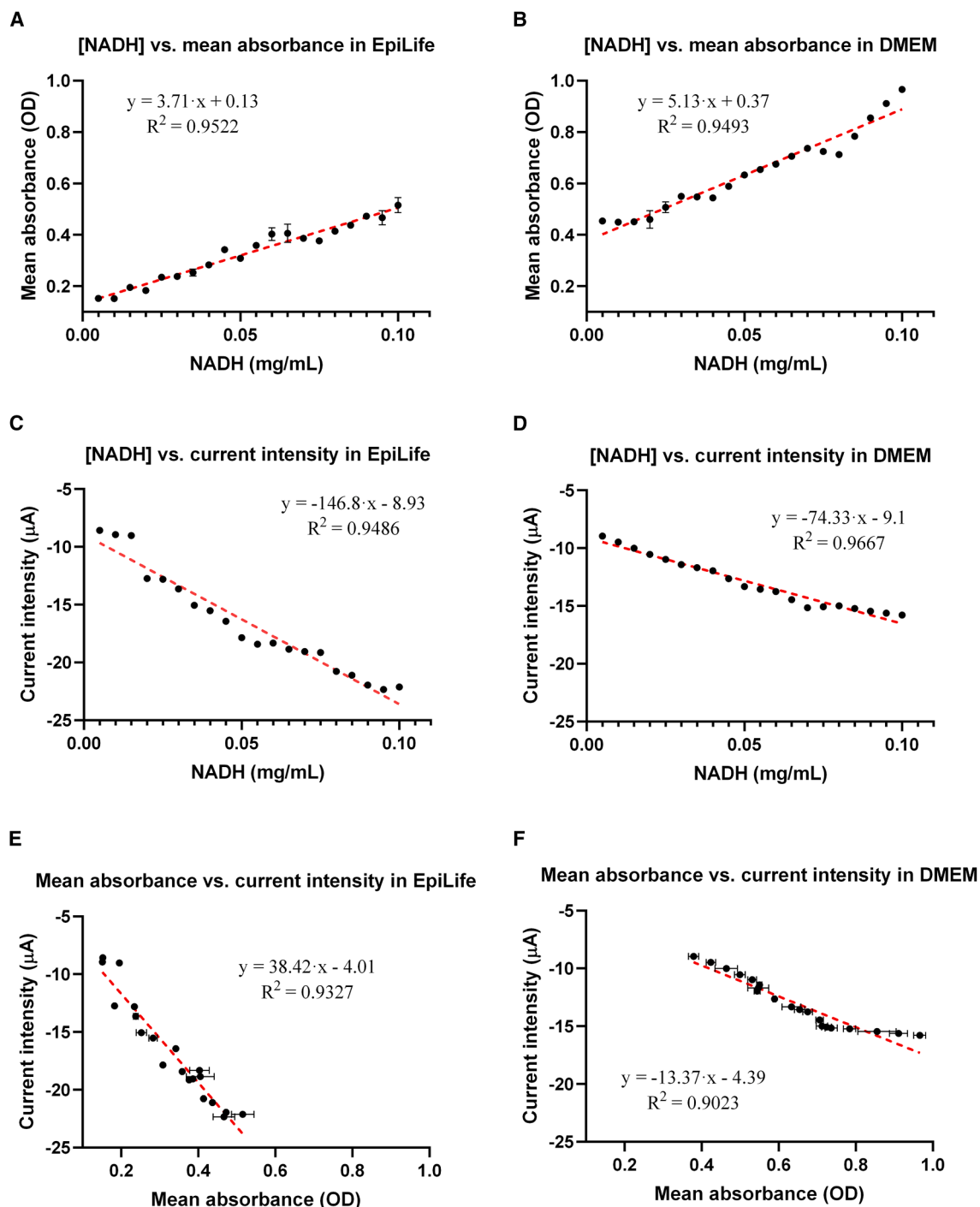


Figure 3. Correlation analysis of MTS reduction by NADH in EpiLife and DMEM culture media

(A) Mean absorbance (OD) as a function of NADH concentration in EpiLife medium (conventional spectrophotometric method, $R^2 = 0.9522$).
 (B) Electrochemical current intensity peak as a function of NADH concentration in EpiLife medium (proposed electrochemical method, $R^2 = 0.9493$).
 (C) Mean absorbance (OD) as a function of NADH concentration in DMEM medium (conventional spectrophotometric method, $R^2 = 0.9486$).
 (D) Electrochemical current intensity peak as a function of NADH concentration in DMEM medium (proposed electrochemical method, $R^2 = 0.9667$).
 (E) Correlation between mean absorbance (OD) and electrochemical current intensity peak in EpiLife medium ($R^2 = 0.9327$).
 (F) Correlation between mean absorbance (OD) and electrochemical current intensity peak in DMEM medium ($R^2 = 0.9023$). Data are represented as mean $\pm$ standard deviation (SD) for $n = 7$ replicates.

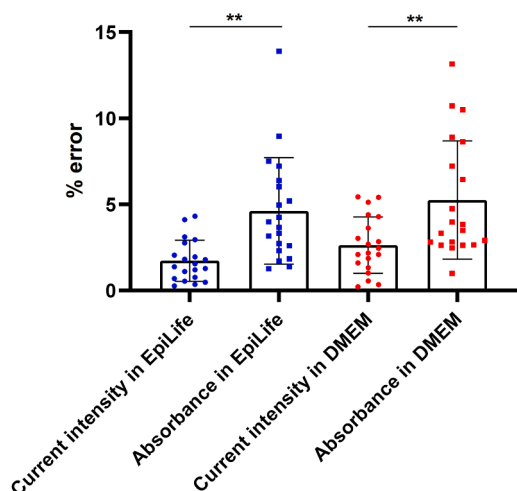


Figure 4. Comparison of the percentage error in MTS reduction measurements obtained via the electrochemical signal versus absorbance readings in EpiLife and DMEM culture media

Bar graphs represent the mean percentage error, with individual data points overlaid. Error bars indicate standard deviation. The electrochemical measurements consistently demonstrated significantly lower variability and greater precision compared to the conventional spectrophotometric method. Statistical significance is denoted by ** ($p < 0.01$, derived from one-way ANOVA).

following chemical exposure. This experiment aimed to validate the proposed electrochemical method as a reliable procedure for chemical toxicity determination, aligning with established guidelines for assessing adverse outcomes such as tissue damage. Chemicals utilized in this study were selected based on their GHS classifications: 5% KOH and 1-decanol are categorized as irritants (Category 2), while isopropanol and 1-bromo-4-chlorobutane are classified as “No Category” (non-irritants). PBS served as negative control and 5% SDS as positive control.

MTS reduction measurements, obtained from both the electrochemical sensor and the conventional spectrophotometric method, showed consistent trends in both primary hDFs and immortalized N/TERT-1 human keratinocytes after 3-min chemical exposure and a 15-min recovery period. As shown in Figure 6, both methods clearly demonstrated a significant reduction in MTS signal for the known irritants (SDS 5% aq., KOH 5% aq., and 1-decanol), indicating cell damage. On the contrary, the “No Category” chemicals (isopropanol, 1-bromo-4-chlorobutane) generally showed higher MTS signals, reflecting less cellular toxicity. These preliminary trends indicate qualitative agreement between the two methods, identifying the toxic potential of the tested chemicals.

Cell viability percentages were subsequently calculated by normalizing the MTS reduction signals to the negative control (PBS), which was set as 100% viability, and the positive control (SDS 5% aq.) as 0% viability. A direct comparison of the calculated viabilities (Figure 7) revealed no statistically significant differences between the electrochemical and spectrophotometric methods for most chemicals and cell lines. The only exception was observed for 5% KOH exposure in hDFs, where a statistically significant difference was found between the two methods ($p < 0.01$).

RHE viability after chemical exposure

To further validate the clinical and regulatory relevance of the electrochemical sensor for skin toxicity assessment, its performance was evaluated using RHE models. RHE models are advanced *in vitro* tissue constructs that closely mimic the physiological and morphological characteristics of human skin, making them highly valuable alternatives to animal testing for irritation and corrosion studies.⁴²

As demonstrated in Figure 8, both the electrochemical sensor and the conventional spectrophotometric method consistently detected changes in RHE viability after exposure to various chemicals. Similar to the monolayer cell cultures, RHE viability was calculated relative to the negative control (PBS), set as 100%. The positive control, 5% aqueous SDS, induced near-complete cell death, with both methods showing viability close to 0%.

For irritant chemicals such as 5% KOH, HCl, 1-decanol, and sulfuric acid, both detection methods indicated a significant reduction in RHE viability compared to the negative control. In contrast, chemicals previously classified as “No Category” or milder irritants, such as 1-bromo-4-chlorobutane, and isopropanol, generally showed higher residual viability, indicating their reduced toxic potential to the RHE model⁴³ and in agreement with test guidelines.

Crucially, a direct comparison of the viability percentages obtained by the electrochemical and spectrophotometric methods for RHE models revealed a high degree of agreement. For all tested chemicals, no statistically significant differences were observed between the two measurement approaches ($p < 0.001$ for SDS 5% aq., and no significance for the others).

While the inherent biological variability of the RHE models increased the standard deviation in both methods (Figures 5, 6, and 7), the electrochemical sensor consistently provided a robust correlation with the cell viability state, matching the reliability of the traditional spectrophotometric assay while offering the advantage of being non-destructive.

DISCUSSION

The development of reliable analytical tools for complex *in vitro* models requires overcoming the limitations of traditional optical assays. The results presented in this study demonstrate that the proposed electrochemical platform successfully decouples viability measurements from optical constraints.

A significant advantage of the developed electrochemical sensor is its ability to eliminate potential interferences arising from cell culture media components, such as DMEM or EpiLife, confirming that, unlike conventional colorimetric assays, which can be susceptible to background absorbance from colored media,¹³ this methodology shows a clear stability across various culture media tested.

The strong linear correlation observed between NADH concentration, and the electrochemical signal ($R^2 > 0.90$) validates the sensor's ability to quantitatively monitor metabolic fluctuations. The precise and sensitive detection of this negative peak current provides a robust and direct measure of reducing equivalents, establishing the basis for accurate cell viability assessment in subsequent biological applications, surpassing the limitations often associated with traditional colorimetric methods.⁴⁴

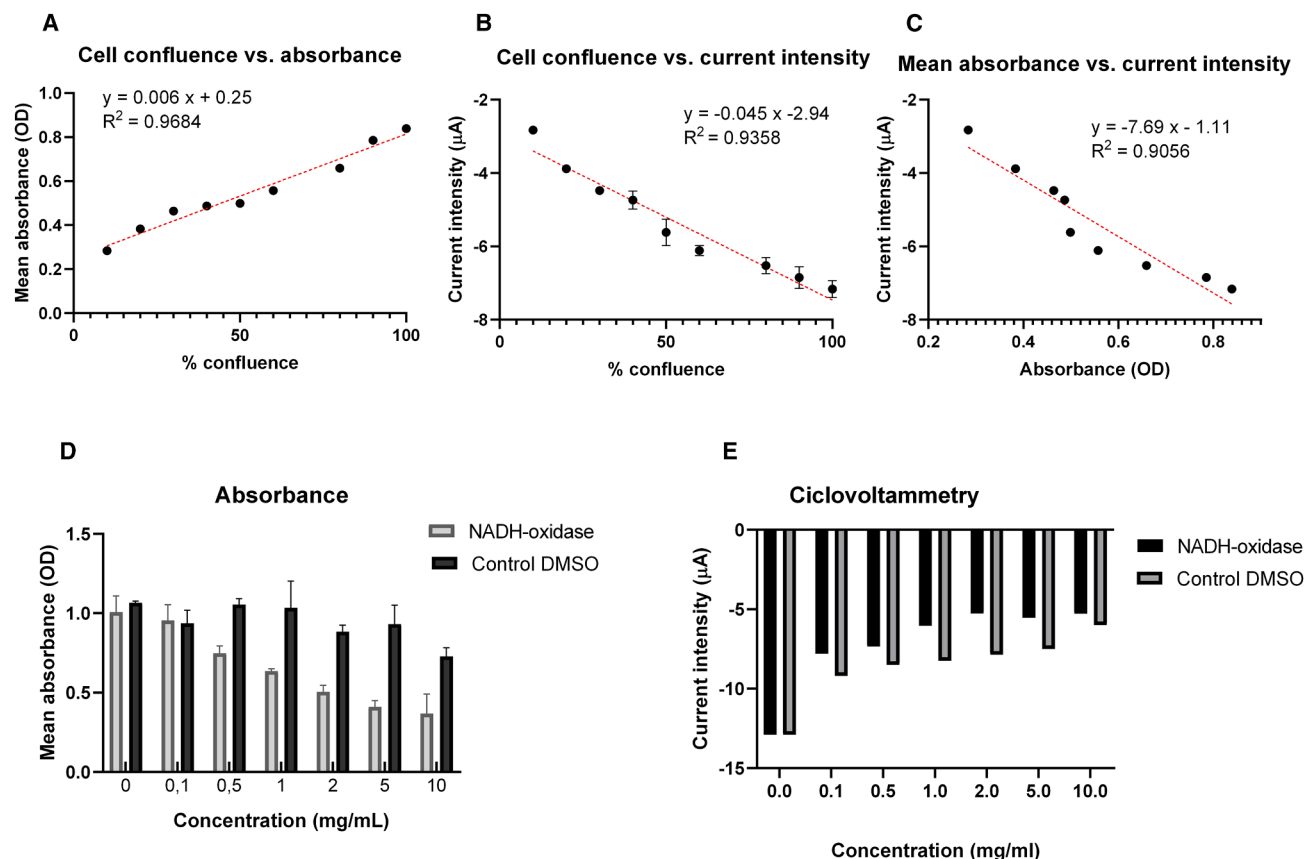


Figure 5. Quantitative assessment of MTS reduction in human dermal fibroblast (hDF) cell confluency after a 3-h incubation with the MTS reagent

(A) Linear correlation between cell confluency percentage and mean MTS absorbance (OD) as measured by the conventional spectrophotometric method ($R^2 = 0.9684$).

(B) Linear correlation between cell confluency percentage and electrochemical current intensity peak measured by the proposed sensor ($R^2 = 0.9358$).

(C) Correlation between the electrochemical current intensity peak and the spectrophotometric mean absorbance (OD) ($R^2 = 0.9056$).

(D) Electrochemical current intensity in cells treated with NADH-oxidase and its corresponding control with DMSO.

(E) Electrochemical signal response in cells treated with NADH-oxidase and its corresponding control (DMSO). The significant reduction in the peak current upon NADH inhibition confirms the metabolic origin of the formazan-derived signal. All data are represented as mean $\pm$ standard deviation (SD) for $n = 7$ replicates.

The high coefficients of determination ($R^2 > 0.90$ for both EpiLife and DMEM) confirm the robust dependence response of MTS reduction to NADH concentration for both measurement techniques, thus highlighting the sensitivity and reliability of the electrochemical approach.^{18,45} Unlike traditional spectrophotometric assays that can be limited by chemical interference or optical interferences and path-length dependency due to the Beer-Lambert Law,³⁵ the proposed electrochemical platform relies on surface-area-dependent redox reactions. This approach provides a clear quantitative readout with a sensitivity (LOD: 0.0046 mg/mL) that matches the requirements for high-throughput screening in bioengineering applications.

The strong reduction in measurement variability is essential for ensuring the reliability of results and enhancing confidence in subsequent cell metabolic activity measurements. These findings collectively highlight enhanced reliability and precision of the electrochemical method for MTS reduction assessment across different relevant biological media.^{11,46}

Importantly, this correlation held true in biological environments containing living cells (hDFs), as evidenced by the linearity with cell confluency. While cells are known to secrete various biomolecules, including growth factors, hormones, and soluble signaling molecules,⁴⁷ which could potentially interfere with electrochemical measurements, these results demonstrated no apparent disturbance in the sensor's readings. However, the ability to distinguish between varying cell densities with high precision further reinforces the specificity and robustness of the electrochemical method for monitoring cell proliferation dynamics in real-time.

While validated here using skin-derived cells, fibroblasts and keratinocytes, the underlying principle of the sensor is universal to mammalian metabolism. The results indicated that the electrochemical sensor provided comparable quantitative viability assessments to the established colorimetric assay. Besides, the successful discrimination between irritants and non-irritants in both cell types suggests that this platform is for reliable,

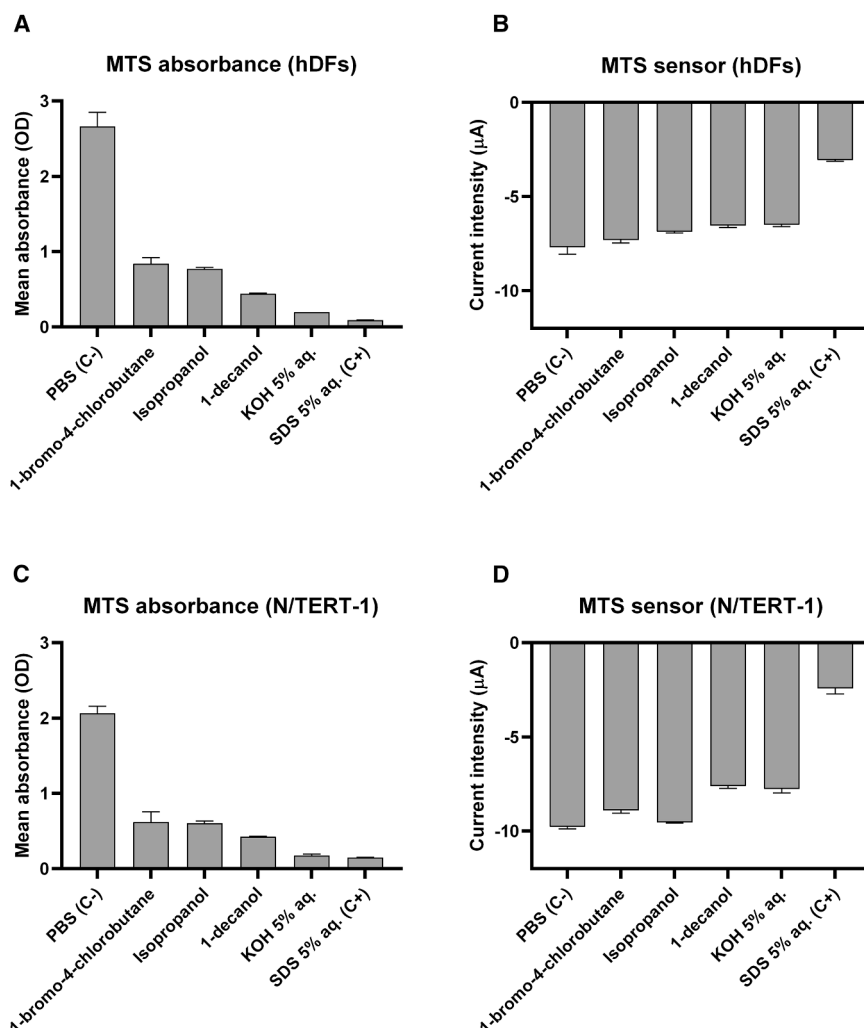


Figure 6. MTS reduction measurements in hDFs and N/TERT-1 keratinocytes after chemical exposure after a 3-h incubation with the MTS reagent

(A) Electrochemical current intensity in hDFs and (B) corresponding MTS absorbance in hDFs. (C) Electrochemical current intensity in N/TERT-1 keratinocytes and (D) corresponding MTS absorbance in N/TERT-1 keratinocytes. Chemicals tested include PBS (negative control, C-), 1-bromo-4-chlorobutane, isopropanol, 1-decanol, 5% aq. KOH, and 5% aq. SDS (positive control, C+). Data are represented as mean $\pm$ standard deviation (SD).

with endpoint structural assessments on a single RHE, thereby reducing experimental variability and the number of required tissue models.

By eliminating optical interference, this technology enables the analysis of wider range of compounds, where traditional colorimetric assays often fail or require destructive processing steps. Since the measurement is performed on the culture supernatant, the 3D reconstructed tissue remains functionally and structurally intact. This characteristic is crucial for advanced toxicological studies, as it enables subsequent analyses, such as histology or cytokine quantification, on the same tissue sample, thereby maximizing the data obtained per experiment and reducing the number of models required (Table S1). Moreover, this capability enables the integration of this sensor into automated, high-throughput screening

consistent, and highly predictive cell viability regardless of their origin.

The specific statistical agreement between the electrochemical and optical methods in the chemical exposure panel confirms that the electrochemical readout is equivalent to the industrial gold standard, but with the added benefits of rapid readout and potential for miniaturization.

The most critical finding of this work is the validation of the sensor in RHE models. 3D tissues represent the most challenging matrix for viability assays due to their thickness and opacity. The fact that the electrochemical sensor matched the spectrophotometric results with no statistically significant differences proves that the method is unaffected by the complex tissue architecture. This strong concordance validates the electrochemical capability of the sensor to accurately and reliably assess the viability of complex 3D human skin models,^{48–51} providing a robust and ethical alternative for *in vitro* toxicology and drug safety evaluation in tissue engineering. As demonstrated by the histological analysis in Figure 8, the tissue architecture remains intact post-measurement. This allows for a multi-parametric characterization, enabling researchers to correlate real-time metabolic data

platforms for drug development and safety testing, aligning with the global shift toward animal-free testing methodologies.

In conclusion, this study establishes a robust electrochemical platform for the precise quantification of cell viability in complex *in vitro* models. By coupling the metabolic specificity of MTS with electrochemical detection, we have successfully eliminated the dependence on optical transparency, overcoming the sensitivity and interference limitations of traditional colorimetric assays. The validation on reconstructed human epidermis (RHE), a challenging, stratified tissue, demonstrates the sensor's transversal applicability, proving it matches regulatory gold standards while offering superior precision.

However, beyond skin irritation, this technology represents a versatile tool for the broader field of toxicology and drug development. Its ability to function in opaque matrices and its potential for miniaturization make it an ideal candidate for integration into automated, and scalable for high-throughput screening pipelines. Ultimately, this platform provides a reliable, quantifiable solution that supports the global regulatory shift toward animal-free testing, enabling the wider adoption of predictive 3D tissue models in biomedical research.

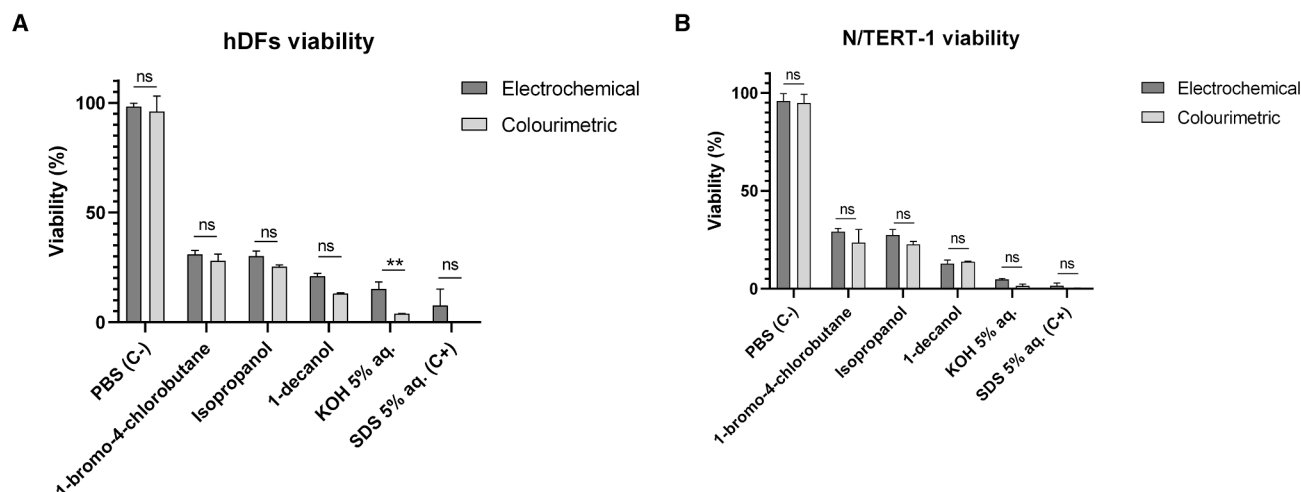


Figure 7. Percentage cell viability calculated from electrochemical and spectrophotometric MTS measurements following chemical exposure after a 3-h incubation with the MTS reagent

(A) Viability in hDFs and (B) viability in N/TERT-1 keratinocytes. Viability was normalized with PBS (negative control) set at 100% and SDS 5% aq. (positive control) at 0%. No statistically significant differences (ns) were observed between the electrochemical and colorimetric methods for most conditions, except for 5% KOH in hDFs, which showed statistical significance (** $p < 0.01$, as determined by two-way ANOVA). Data are represented as mean $\pm$ standard deviation (SD) for $n = 7$ replicates.

Limitations of the study

Although the developed electrochemical platform offers high precision and successfully eliminates optical interferences in RHE models, some constraints remain. The current configuration relies on manual collection of the culture supernatant after

a 4-h incubation period with the MTS reagent, meaning it operates as a standardized endpoint evaluation rather than a fully automated, continuous real-time monitoring system. Additionally, while the metabolic principle of the sensor is transversal to all mammalian cells, the system has been validated exclusively

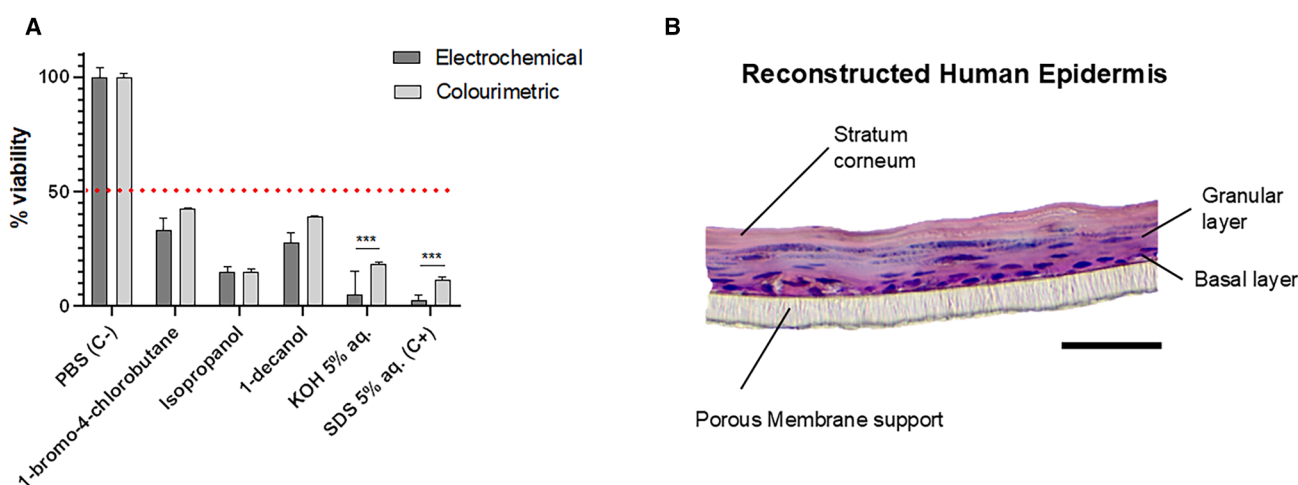


Figure 8. Evaluation of the methodology for a complex epithelial tissue

(A) Percentage viability of reconstructed human epidermis (RHE) models after chemical exposure, calculated from both electrochemical and spectrophotometric MTS measurements. Viability was normalized to the negative control (PBS), set as 100%. Chemicals tested include PBS (C-), 5% aq. SDS (C+), 1-bromo-4-chlorobutane, 1-decanol, HCl, sulfuric acid, 5% aq. KOH, and isopropanol. Data are represented as mean $\pm$ standard deviation (SD) for $n = 3$ replicates. Red dashed line indicates IC50 as 50% of viability for chemical corrosive and irritation assessment. Statistical significance for comparison between electrochemical and colorimetric methods for each chemical: no significance (ns), p value < 0.001 (***).

(B) Histological characterization of the RHE model. Representative H&E stained section of the reconstructed human epidermis (RHE) used for electrochemical validation. The image confirms the presence of a fully stratified epithelium, including a prominent stratum corneum (SC), granular layer, and basal layer. This complex architecture demonstrates the robustness of the electrochemical MTS assay in monitoring viability within a highly organized 3D barrier model. Scale bars, 50 μ m.

within a specific skin toxicity panel and validation in other tissue models could be considered in future studies.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and additional data should be directed to and will be fulfilled by the lead contact, Diego Velasco (divelasc@ing.uc3m.es).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- Data reported in this paper will be shared by the [lead contact](#) upon request.
- This paper does not report any original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

ACKNOWLEDGMENTS

The authors want to acknowledge the valuable help from Jos Smits from Radboud University (Nijmegen, The Netherlands) to establish the proper conditions for the RHE establishment. This article has been partially funded through the R&D activities program with reference TEC-2024/BIO-98 and acronym NeuroBiopieltec-CM, granted by the Community of Madrid through the General Directorate of Research and Technological Innovation through Order 5696/2024, of December 10, of the Minister of Education, Science and Universities. The work of Maarten van Os has been supported by the European Union's Horizon Europe research and innovation program under the Marie Skłodowska-Curie Doctoral Networks grant agreement no. 101169419, TeralBs. Funding for APC: Universidad Carlos III de Madrid (agreement CRUE-Madroño 2025). We also acknowledge Salas Blancas of Bioengineering facility core of Universidad Carlos III de Madrid, which housed most of our experimental activities.

AUTHOR CONTRIBUTIONS

Conceptualization, P.A., L.V., and D.V.; methodology, I.R., S.N., M.v.O., and L.V.; investigation, I.R., S.N., M.v.O., and L.V.; writing – original draft, I.R., S.N., and L.V.; writing – review and editing, P.A., L.V., and D.V.; funding acquisition, P.A. and D.V.; resources, P.A. and D.V.; supervision, P.A. and D.V.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

The authors used Gemini, an AI-assisted tool, to improve spelling, grammar, clarity, and readability during manuscript preparation. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- [KEY RESOURCES TABLE](#)
- [EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS](#)
 - Human dermal fibroblast cell cultures
 - N/TERT-1 keratinocyte cell cultures
 - Reconstructed Human Epidermis cultures
- [METHOD DETAILS](#)

- Electrochemical measurements
- Validation of the MTS electrochemical signal
- NADH production inhibition
- Histological processing
- MTS assay
- Cell confluency and MTS electrochemical signal correlation
- Cell viability assessment after chemical exposure
- Skin irritation assessment after chemical exposure

● QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.isci.2026.117118>.

Received: January 2, 2026

Revised: May 28, 2026

Accepted: July 23, 2026

REFERENCES

1. Vinardell, M.P., and Mitjans, M. (enero de 2008). Alternative Methods for Eye and Skin Irritation Tests: An Overview. *J. Pharm. Sci.* 97, 46–59. <https://doi.org/10.1002/jps.21088>.
2. Chacón, M., Vázquez, N., Persinal-Medina, M., Alonso-Alonso, S., Pevida, M., Llamas, S., Baamonde, B., Quiros, L., Merayo-Lloves, J., and Meana, Á. (2020). Development of an in-house reconstructed human epidermis model as an alternative method in skin corrosion assessment. *Toxicol. Vitro* 65, 104779. <https://doi.org/10.1016/j.tiv.2020.104779>.
3. Chacón, M., Vázquez, N., Alonso-Alonso, S., Llamas, S., Pevida, M., Alcalde, I., Merayo-Lloves, J., and Meana, Á. (de 2024). Evaluating the QileX-RHE skin corrosion test for chemical subcategorization in accordance with OECD TG 431. *Toxicology in Vitro* 98, 105816. <https://doi.org/10.1016/j.tiv.2024.105816>.
4. Pistollato, F., Madia, F., Corvi, R., Munn, S., Grignard, E., Paini, A., Worth, A., Bal-Price, A., Prieto, P., Casati, S., et al. (2021). Current EU regulatory requirements for the assessment of chemicals and cosmetic products: challenges and opportunities for introducing new approach methodologies. *Arch. Toxicol.* 95, 1867–1897. <https://doi.org/10.1007/s00204-021-03034-y>.
5. Losada-Fernández, I., San Martín, A., Moreno-Nombela, S., Suárez-Cabrera, L., Valencia, L., Pérez-Aciego, P., and Velasco, D. (2025). In Vitro Skin Models for Skin Sensitisation: Challenges and Future Directions. *Cosmetics* 12, 173. <https://doi.org/10.3390/cosmetics12040173>.
6. Xie, R., Pal, V., Yu, Y., Lu, X., Gao, M., Liang, S., Huang, M., Peng, W., and Ozbolat, I.T. (2024). A comprehensive review on 3D tissue models: Bio-fabrication technologies and preclinical applications. *Biomaterials* 304, 122408. <https://doi.org/10.1016/j.biomaterials.2023.122408>.
7. Pennarossa, G., Arcuri, S., De Iorio, T., Gandolfi, F., and Brevini, T.A.L. (2021). Current Advances in 3D Tissue and Organ Reconstruction. *Int. J. Mol. Sci.* 22, 830. <https://doi.org/10.3390/ijms22020830>.
8. Delrue, N., Sachana, M., Sakuratani, Y., Gourmelon, A., Leinla, E., and Diderich, R. (2016). The Adverse Outcome Pathway Concept: A Basis for Developing Regulatory Decision-making Tools. *Altern. Lab. Anim.* 44, 417–429. <https://doi.org/10.1177/026119291604400504>.
9. Poumay, Y., Dupont, F., Marcoux, S., Leclercq-Smekens, M., Hérin, M., and Coquette, A. (2004). A simple reconstructed human epidermis: preparation of the culture model and utilization in in vitro studies. *Arch. Dermatol. Res.* 296, 203–211. <https://doi.org/10.1007/s00403-004-0507-y>.
10. Huet, O., Petit, J.M., Ratinaud, M.H., and Julien, R. (1992). NADH-dependent dehydrogenase activity estimation by flow cytometric analysis of 3-(4,5-dimethylthiazolyl-2-yl)-2,5-diphenyltetrazolium bromide (MTT) reduction. *Cytometry* 13, 532–539. <https://doi.org/10.1002/cyto.990130513>.

11. Ghasemi, M., Turnbull, T., Sebastian, S., and Kempson, I. (noviembre de 2021). The MTT Assay: Utility, Limitations, Pitfalls, and Interpretation in Bulk and Single-Cell Analysis. *Int. J. Mol. Sci.* 22, 12827. <https://doi.org/10.3390/ijms222312827>.
12. Luis, C., Castaño-Guerrero, Y., Soares, R., Sales, G., and Fernandes, R. (abril de 2019). Avoiding the Interference of Doxorubicin with MTT Measurements on the MCF-7 Breast Cancer Cell Line. *Methods Protoc.* 2, 29. <https://doi.org/10.3390/mps2020029>.
13. Huang, K.T., Chen, Y.H., and Walker, A.M. (2004). Inaccuracies in MTS Assays: Major Distorting Effects of Medium, Serum Albumin, and Fatty Acids. *Biotechniques* 37, 406–412. <https://doi.org/10.2144/04373ST05>.
14. Choi, J., Kim, H., Choi, J., Oh, S.M., Park, J., and Park, K. (2014). Skin corrosion and irritation test of sunscreen nanoparticles using reconstructed 3D human skin model. *Environ. Health Toxicol.* 29, e2014004. <https://doi.org/10.5620/eht.2014.29.e2014004>.
15. Gaviria Agudelo, C., Becerra, N.Y., Vergara, J.D., Correa, L.A., Estrada, S., and Restrepo, L.M. (marzo de 2020). Dermo-epidermal organotypic cultures for in vitro evaluation of skin irritation and corrosion. *Toxicol. Vitro* 63, 104657. <https://doi.org/10.1016/j.tiv.2019.104657>.
16. Carlin, M., Garrido, M., Sosa, S., Tubaro, A., Prato, M., and Pelin, M. (2023). *In vitro* assessment of skin irritation and corrosion properties of graphene-related materials on a 3D epidermis. *Nanoscale* 15, 14423–14438. <https://doi.org/10.1039/D3NR03081D>.
17. Arab-Bafrani, Z., Shahbazi-Gahrouei, D., Abbasian, M., and Fesharaki, M. (2016). Multiple MTS Assay as the Alternative Method to Determine Survival Fraction of the Irradiated HT-29 Colon Cancer Cells. *J. Med. Signals Sens.* 6, 112–116.
18. Dunigan, D.D., Waters, S.B., and Owen, T.C. (noviembre de 1995). Aqueous soluble tetrazolium/formazan MTS as an indicator of NADH- and NADPH-dependent dehydrogenase activity. *Biotechniques* 19, 640–649. <https://www.researchgate.net/publication/14425880>.
19. Wright, A., and Mowrey-McKee, M. (2005). COMPARATIVE CYTOTOXICITY POTENTIAL OF SOFT CONTACT LENS CARE PRODUCTS. *Cutaneous and Ocular Toxicology.* enero de 24, 53–64. <https://doi.org/10.1081/CUS-200046191>.
20. Tsai, T.Y., Chen, T.C., Wang, I.J., Yeh, C.Y., Su, M.J., Chen, R.H., Tsai, T.H., and Hu, F.R. (2015). The Effect of Resveratrol on Protecting Corneal Epithelial Cells from Cytotoxicity Caused by Moxifloxacin and Benzalkonium Chloride. *Investig. Ophthalmol. Vis. Sci.* 56, 1575–1584. <https://doi.org/10.1167/iov.14-15708>.
21. Pham, D.D.D., Jenčová, V., Kaňuchová, M., Bayram, J., Grossová, I., Šuca, H., Urban, L., Havlíčková, K., Novotný, V., Mikeš, P., et al. (2021). Novel lipophosphonoxin-loaded polycaprolactone electrospun nanofiber dressing reduces *Staphylococcus aureus* induced wound infection in mice. *Scientific Reports.* septiembre de 11, 17688. <https://doi.org/10.1038/s41598-021-96980-7>.
22. Veit, J.G.S., Weidow, M., and Serban, M.A. (2023). A versatile, bio-engineered skin reconstruction device designed for use in austere environments. *Front. Bioeng. Biotechnol.* 11, 1208322. <https://doi.org/10.3389/fbioe.2023.1208322>.
23. Faller, C., Bracher, M., Dami, N., and Roguet, R. (2002). Predictive ability of reconstructed human epidermis equivalents for the assessment of skin irritation of cosmetics. *Toxicol. Vitro* 16, 557–572. [https://doi.org/10.1016/S0887-2333\(02\)00053-X](https://doi.org/10.1016/S0887-2333(02)00053-X). www.elsevier.com/locate/toxinvit.
24. Rodrigues, D., Barbosa, A.I., Rebelo, R., Kwon, I.K., Reis, R.L., and Correo, V.M. (2020). Skin-Integrated Wearable Systems and Implantable Biosensors: A Comprehensive Review. *Biosensors* 10, 79. <https://doi.org/10.3390/bios10070079>.
25. Lee, E.K., Kim, M.K., and Lee, C.H. (2019). Skin-Mountable Biosensors and Therapeutics: A Review. *Annu. Rev. Biomed. Eng.* 21, 299–323. <https://doi.org/10.1146/annurev-bioeng-060418-052315>.
26. Ullah, S., Hamade, F., Bubniene, U., Engblom, J., Ramanavicius, A., Ramanaviciene, A., and Ruzgas, T. (2018). In-vitro model for assessing glucose diffusion through skin. *Biosens. Bioelectron.* 110, 175–179. <https://doi.org/10.1016/j.bios.2018.03.039>.
27. Rothbauer, M., Strauss, J., Reihls, E.I., Heidenberger, J., Alphonsus, J., Gowers, S., Chen, Y.C., McLeod, J., Toegel, S., Ertl, P., et al. (2025). Integration of glucose and lactate biosensors into human cartilage-on-a-chip models for long-term monitoring of metabolic shifts in osteoarthritis. *Sensor. Actuator. B Chem.* 427, 137123. <https://doi.org/10.1016/j.snb.2024.137123>.
28. Weber, J., Kumar, A., Kumar, A., and Bhansali, S. (2006). Novel lactate and pH biosensor for skin and sweat analysis based on single walled carbon nanotubes. *Sensor. Actuator. B Chem.* 117, 308–313. <https://doi.org/10.1016/j.snb.2005.12.025>.
29. van den Brink, N.J.M., Pardow, F., Meesters, L.D., van Vlijmen-Willems, I., Rodijk-Olthuis, D., Niehues, H., Jansen, P.A.M., Roelofs, S.H., Brewer, M.G., van den Bogaard, E.H., et al. (2024). Electrical Impedance Spectroscopy Quantifies Skin Barrier Function in Organotypic In Vitro Epidermis Models. *J. Invest. Dermatol.* 144, 2488–2500.e4. <https://doi.org/10.1016/j.jid.2024.03.038>.
30. Chacón, M., Vázquez, N., Alonso-Alonso, S., Persinal-Medina, M., Llamas, S., Pevida, M., Alcalde, I., Merayo-Llodes, J., and Meana, Á. (2023). Improved Tool for Predicting Skin Irritation on Reconstructed Human Epidermis Models Based on Electrochemical Impedance Spectroscopy. *Biosensors* 13, 162. <https://doi.org/10.3390/bios13020162>.
31. Ceccarelli, M.C., Lefevre, M.C., Marino, A., Pignatelli, F., Krukiewicz, K., Battaglini, M., and Ciofani, G. (2024). Real-time monitoring of a 3D blood-brain barrier model maturation and integrity with a sensorized microfluidic device. *Lab Chip* 24, 5085–5100. <https://doi.org/10.1039/D4LC00633J>.
32. Patra, S., Kundu, D., Gogoi, M., and editores. (2023). *Enzyme-based Biosensors: Recent Advances and Applications in Healthcare* (Springer Nature). <https://doi.org/10.1007/978-981-15-6982-1>. <https://link.springer.com/10.1007/978-981-15-6982-1>.
33. Maryam, H.K., and Abdullah, F.H. (2025). Electrochemical Biosensors for Real-Time Oxidative Stress Monitoring in Saliva. *J. Eng. Sci. Technol. Trends.* 2. <https://doi.org/10.64060/jestt.v2i2.3>.
34. Koo, K.M., Kim, C.D., Kim, H., Cho, Y.W., Suhito, I.R., and Kim, T.H. (2023). Extracellularly Detectable Electrochemical Signals of Living Cells Originate from Metabolic Reactions. *Adv. Sci.* 10, 2207084. <https://doi.org/10.1002/advs.202207084>.
35. Pour, S.R.S., Calabria, D., Emamiain, A., Lazzarini, E., Pace, A., Guardigli, M., Zangheri, M., and Mirasoli, M. (2023). Electrochemical vs. Optical Biosensors for Point-of-Care Applications: A Critical Review. *Chemosensors* 11, 546. <https://doi.org/10.3390/chemosensors11100546>.
36. Batool, S., Javaid, S., Javed, H., Laiba, A., Shahid, I., Khan, M., and Muhammad, A. (7 de septiembre de 2022). Addressing artifacts of colorimetric anticancer assays for plant-based drug development. *Med. Oncol.* 39, 198. <https://doi.org/10.1007/s12032-022-01791-z>.
37. Batchelor-McAuley, C., Kätelhön, E., Barnes, E.O., Compton, R.G., LaBorda, E., and Molina, A. (2015). Recent Advances in Voltammetry. *ChemistryOpen* 4, 224–260. <https://doi.org/10.1002/open.201500042>.
38. Berridge, M.V., Herst, P.M., and Tan, A.S. (2005). Tetrazolium dyes as tools in cell biology: New insights into their cellular reduction. *Biotechnol. Annu. Rev.* 11, 127–152. [https://doi.org/10.1016/S1387-2656\(05\)11004-7](https://doi.org/10.1016/S1387-2656(05)11004-7).
39. Umamoto, K. (1989). Electrochemical Studies of the Reduction Mechanism of Tetrazolium Salts and Formazans. *Bull. Chem. Soc. Jpn.* 62, 3783–3789. <https://doi.org/10.1246/bcsj.62.3783>.
40. Shrivastava, A., and Gupta, V. (2011). Methods for the determination of limit of detection and limit of quantitation of the analytical methods. *Chron. Young Sci.* 2, 21. <https://doi.org/10.4103/2229-5186.79345>.
41. Lavín, Á., Vicente, J.D., Holgado, M., Laguna, M.F., Casquel, R., Santamaría, B., Maigler, M.V., Hernández, A.L., and Ramírez, Y. (2018). On the Determination of Uncertainty and Limit of Detection in Label-Free Biosensors. *Sensors (Basel)* 18, 2038. <https://doi.org/10.3390/s18072038>.

42. Kidd, D.A., Johnson, M., and Clements, J. (octubre de 2007). Development of an in vitro corrosion/irritation prediction assay using the EpiDerm™ skin model. *Toxicol. Vitro* 21, 1292–1297. <https://doi.org/10.1016/j.tiv.2007.08.018>.
43. (2015). Performance Standards for the Assessment of Proposed Similar or Modified In Vitro Reconstructed Human Epidermis (RhE) Test Methods for Skin Irritation Testing as Described in TG 439 (OECD). https://www.oecd.org/en/publications/performance-standards-for-the-assessment-of-proposed-similar-or-modified-in-vitro-reconstructed-human-epidermis-rhe-test-methods-for-skin-irritation-testing-as-described-in-tg-439_745487b9-en.html.
44. Hsieh, M.S., Wu, T.G., Su, C.S., Cheng, W.J., Ozbek, N., Tsai, K.Y., and Lin, C.Y. (2011). Comparison of an electrochemical biosensor with optical devices for hemoglobin measurement in human whole blood samples. *Clin. Chim. Acta* 412, 2150–2156. <https://doi.org/10.1016/j.cca.2011.07.026>.
45. Oritani, T., Fukuhara, N., Okajima, T., Kitamura, F., and Ohsaka, T. (2004). Electrochemical and spectroscopic studies on electron-transfer reaction between novel water-soluble tetrazolium salts and a superoxide ion. *Inorganica Chim. Acta* 357, 436–442. <https://doi.org/10.1016/j.ica.2003.05.007>.
46. Vega-Avila, E., and Pugsley, M.K. (2011). An overview of colorimetric assay methods used to assess survival or proliferation of mammalian cells. *Proc. West. Pharmacol. Soc.* 54, 10–14.
47. Dowling, P., and Clynes, M. (2011). Conditioned media from cell lines: A complementary model to clinical specimens for the discovery of disease-specific biomarkers. *Proteomics* 11, 794–804. <https://doi.org/10.1002/pmic.201000530>.
48. Smits, J.P.H., Niehues, H., Rikken, G., Van Vlijmen-Willems, I.M.J.J., Van De Zande, G.W.H.J.F., Zeeuwen, P.L.J.M., Schalkwijk, J., and van den Bogaard, E.H. (2017). Immortalized N/TERT keratinocytes as an alternative cell source in 3D human epidermal models. *Sci. Rep.* 7, 11838. <https://doi.org/10.1038/s41598-017-12041-y>.
49. Seo, M.D., Kang, T.J., Lee, C.H., Lee, A.Y., and Noh, M. (2012). HaCaT Keratinocytes and Primary Epidermal Keratinocytes Have Different Transcriptional Profiles of Cornified Envelope-Associated Genes to T Helper Cell Cytokines. *Biomol. Ther.* 20, 171–176. <https://doi.org/10.4062/bio-molther.2012.20.2.171>.
50. Van Drongelen, V., Danso, M.O., Mulder, A., Mieremet, A., Van Smeden, J., Bouwstra, J.A., and El Ghalbzouri, A. (noviembre de 2014). Barrier Properties of an N/TERT-Based Human Skin Equivalent. *Tissue Eng. Part A* 20, 3041–3049. <https://doi.org/10.1089/ten.tea.2014.0011>.
51. Moran, M.C., Pandya, R.P., Leffler, K.A., Yoshida, T., Beck, L.A., and Brewer, M.G. (2021). Characterization of Human Keratinocyte Cell Lines for Barrier Studies. *JID Innov.* 1, 100018. <https://doi.org/10.1016/j.xjidi.2021.100018>.
52. Dickson, M.A., Hahn, W.C., Ino, Y., Ronfard, V., Wu, J.Y., Weinberg, R.A., Louis, D.N., Li, F.P., and Rheinwald, J.G. (2000). Human keratinocytes that express hTERT and also bypass a p16INK4a-enforced mechanism that limits life span become immortal yet retain normal growth and differentiation characteristics. *Mol. Cell. Biol.* 20, 1436–1447.
53. Régnier, M., Caron, D., Reichert, U., and Schaefer, H. (1992). Reconstructed human epidermis: a model to study in vitro the barrier function of the skin. *Skin Pharm. Physiol.* 5, 49–56.
54. Bajsert, J., De Glas, V., Faway, E., de Rouvroit, C.L., Pérez-Aso, M., Cook, P.W., and Poumay, Y. (2024). Characterization of reconstructed human epidermis in a chemically-defined, animal origin-free cell culture. *JID Innovat.* 4, 100298.
55. Pour, S.R., Calabria, D., Emami Amin, A., Lazzarini, E., Pace, A., Guardigli, M., Zangheri, M., and Mirasoli, M. (2023). Electrochemical vs. optical biosensors for point-of-care applications: a critical review. *Chemosensors* 11, 546.
56. Koo, K.M., Kim, C.D., Kim, H., Cho, Y.W., Suhito, I.R., and Kim, T.H. (2023). Extracellularly detectable electrochemical signals of living cells originate from metabolic reactions. *Adv. Sci.* 10, 2207084.
57. (2015). Performance Standards for the Assessment of Proposed Similar or Modified In Vitro Reconstructed Human Epidermis (RhE) Test Methods for Skin Irritation Testing as described in TG 439 (OECD).
58. El Ghalbzouri, A., Siamari, R., Willemze, R., and Poncet, M. (2008). Leiden reconstructed human epidermal model as a tool for the evaluation of the skin corrosion and irritation potential according to the ECVAM guidelines. *Toxicol. In Vitro* 22, 1311–1320.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
Dulbecco's Modified Eagle Medium (DMEM high glucose with GlutaMAX	Gibco, ThermoFisher	Cat# D6429
Fetal Bovine Serum (FBS)	Sigma Aldrich	Cat# F7524
Penicillin-Streptomycin	ThermoFisher Scientific	Cat# P4333
EpiLife Medium	ThermoFisher Scientific	Cat# MEPI500CA
Human Keratinocyte Growth Serum (HKGS)	ThermoFisher Scientific	Cat# S0015
Nicotinamide adenine dinucleotide (NADH), Grade II, disodium salt	Roche	Cat# 10128023001
BIOMYC™-1 Antibiotic Solution	Sartorius AG, Germany	Cat# 03-036-1C
BIOMYC™-2 Antibiotic Solution	Sartorius AG, Germany	Cat# 03-037-1C
NADH-oxidase	MedChem Express	CAS No.: 2762405-17-2
Bovine Collagen I	Sigma Aldrich	Cat# C9791
CnT-Prime Epithelial 3D Airlift Medium	CELLnTEC	Cat# CnT-PR-3D
Formalin solution, neutral buffered, 10%	Sigma Aldrich	Cat# HT501128-4L
X-free Solvent for histology xylene substitute. IVD in Classe A.	Bio-Optica Milano S.p.A	Cat# 06-1305Q
Absolute ethanol	Supelco	Cat# 1009835000
Alcoholic Eosin Y	Bio-optical, Labolan	Cat# 05-10009/L
Gill I Haematoxylin	Bio-optical, Labolan	Cat# 05-06013/L
Eukitt Mounting medium	Sigma-Aldrich	Cat# 03989-100 ML
2-propanol, ≥99.5%, ACS reagent	Sigma Aldrich	Cat# 190764
Potassium hydroxide, ACS reagent, ≥85%, pellets	Sigma Aldrich	Cat# 221473-25G
1-decanol, 98%	Sigma Aldrich	Cat# 150584-500G
1-bromo-4-chlorobutane, 99%	Sigma Aldrich	Cat# B60800-25G
Hydrochloric acid, 37%, ACS reagent	Sigma Aldrich	Cat# 258148-25 ML
Sulfuric acid solution	Supelco	Cat# 1603151000
Dulbecco's Phosphate Buffered Saline, 5M (10N)	Sigma Aldrich	Cat# D8537
Sodium dodecyl sulfate, ACS reagent, ≥99.0%	Sigma Aldrich	Cat# 436143-25G
Critical commercial assays		
MTS Assay Kit	Promega	Cat# G3582
Experimental models: Cell lines		
Primary human dermal fibroblasts (hDFs)	PromoCell	Cat# C-12300
Immortalized human keratinocytes (N/TERT-1)	Rheinwald laboratory (Harvard Medical School, Boston, USA)	ATCC CVCL_CW92
Software and algorithms		
DropView 8400	Dropsens (Metrohm DropSens).	https://metrohm-dropsens.com/products/software/dropview-8400m-software/
GraphPad Prism	v.8.0.2, GraphPad, Dotmatics, USA	https://www.graphpad.com/
Biorender	BioRender.com	https://biorender.com
Canva professional design software	Canva Pty Ltd, Australia	https://www.canva.com/
Other		
Nunc™ Cell Culture Insert in Carrier Plate Systems (0.4 μm)	ThermoFisher Scientific	Cat# 141002

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Corning® Falcon® 24 Well Culture Plate	ThermoFisher Scientific	Cat# 142475
Corning® 96 Well Clear Polystyrene Microplate	ThermoFisher Scientific	Cat# CLS353075
μStat 300 potentiostat (DRP-STAT300)	Metrohm	N/A
Screen-printed carbon electrodes (SPCE) (DRP-C110)	Metrohm DropSens	N/A
Tissue processor (KD-TS1A)	Kedee Histology Co.	N/A
Microtome (MR 3000)	Histo-Line Laboratories	N/A
Multimode Microplate Reader. Synergy™ HTX	Agilent technologies	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human dermal fibroblast cell cultures

To perform the different biological assays, commercial human primary dermal fibroblasts (hDFs) (Cat# C-12300, PromoCell, Germany) were used. hDFs were cultured in DMEM, 10% FBS, and 1% Ab/Am at 37°C and 5% CO₂. Cell culture medium was changed every 2-3 days, and when 80% confluence was reached, cells were subcultured. All cells were tested for mycoplasma by PCR, and when positive, treated with BIOMYC™-1 and BIOMYC™-2 (Cat# 03-036-1C and Cat# 03-037-1C, respectively, Sartorius AG, Germany) for mycoplasma elimination.

N/TERT-1 keratinocyte cell cultures

To perform the different biological assays, immortalized N/TERT-1 human keratinocytes (ATCC CVCL_CW92) were used. N/TERT-1 keratinocytes were cultured in serum-free EpiLife supplemented with HKGS and 1% Ab/Am at 37°C and 5% CO₂. Culture medium was changed every 2 days and when confluence reached 50-60 % cells were subcultured following protocols from Dickson et al.⁵² and Régner et al.⁵³ All cells were tested for mycoplasma by PCR, and when positive, treated with BIOMYC™-1 and BIOMYC™-2 (Cat# 03-036-1C and Cat# 03-037-1C, respectively, Sartorius AG, Germany) for mycoplasma elimination.

Reconstructed Human Epidermis cultures

To create the RHE, Nunc Cell Culture Inserts in Carrier Plate Systems (0.4 μm pore size) (Cat# 141002, ThermoFisher Scientific) were coated with bovine Collagen I (Cat# C9791, Sigma Aldrich) (0.1 mg/mL) for 1 hour at 37°C, then washed with PBS and air-dried under UV light. N/TERT-1 cells were seeded at a concentration of 4,900 cells/mm². Cells were kept in EpiLife until confluency was reached, and then the inserts were lifted to air-liquid interface (ALI), and culture medium was changed to CnT-Prime 3D Airlift Medium (Cat# CnT-PR-3D, CellnTec): DMEM (60:40) to allow terminal differentiation.⁵⁴ Media was changed every other day, and the epidermis were kept for 10 days to ensure complete differentiation and cornification, changing media every other day.

METHOD DETAILS

Electrochemical measurements

Cyclic voltammetry (CV) response of MTS experiments were performed using a DropSens potentiostat (μStat 300, from Metrohm DropSens, Llanera, Spain) interfaced with DropView 8400 software. All measurements were conducted in a three-electrode configuration using commercial screen-printed carbon electrodes (SPCEs) from Metrohm DropSens (DRP-C110). The electrochemical cell integrated in the strip consisted of a carbon working electrode (WE, 4 mm diameter), a carbon counter electrode (CE), and a silver reference electrode (RE). All potentials reported in this work are referred to this internal Ag reference electrode.

For each experiment, 50 μL of the sample solution was added to the sensor electrode surface, and three consecutive scans were recorded to ensure system stabilization and electrochemical stability of the material. The potential was swept from a starting potential of -0.6 V to a final potential of +0.7 V. The scan rate was set to 50 mV/s. All experiments were performed at room temperature (25°C). After every measurement, the sample was removed and the electrode was washed with Dulbecco's Phosphate Buffered Saline 1X (Cat# D8537, Sigma Aldrich) and carefully dried with absorbent paper.

Following the cell incubation period with the MTS reagent (see Section [MTS assay](#)), the resulting supernatant was collected and measured directly without any prior sample purification or complex pre-processing steps. Furthermore, the screen-printed carbon electrodes (SPCEs) were used without any surface modification or activation.

Validation of the MTS electrochemical signal

Three different culture media, Dulbecco's Modified Eagle Medium (DMEM high glucose with GlutaMAX, Cat# D6429, Gibco, ThermoFisher Scientific) supplemented with a 10% Fetal Bovine Serum (FBS, Cat# F7524, Sigma Aldrich) and 1% antibiotic/

antimycotic (Ab/Am) (Penicillin-Streptomycin, Cat# P4333, ThermoFisher Scientific); and EpiLife Medium (Cat# MEPI500CA, ThermoFisher Scientific) supplemented with Human Keratinocyte Growth Serum (HKGS, Cat# S0015, ThermoFisher Scientific) and 1% Ab/Am, were tested to evaluate their influence on the electrochemical response of the system with cyclic voltammetry.

To further validate the system, control experiments were conducted with each medium and MTS individually to check for potential interferences. For that purpose, nicotinamide adenine dinucleotide (NADH, Cat# 10128023001, Roche) was chosen as the reducing agent of MTS at concentrations ranging from 0.005 to 0.1 mg/mL in intervals of 0.005 mg/mL in DMEM and EpiLife.

These measurements were used to obtain a calibration curve for the sensor relating absorbance and signal intensity, providing detection and saturation limits, and assessing its reproducibility. Additionally, extra measurements were performed with non-reduced MTS, and EpiLife or DMEM as blanks, respectively.

The Limit of Detection (LOD) was calculated according to ICH guidelines using the formula $LOD = 3.3\sigma/S$, where σ is the standard deviation of the residuals ($S_{y,x}$) from the linear regression and S is the slope of the calibration curve.⁵⁵

NADH production inhibition

To evaluate the dependence of the electrochemical signal on cellular metabolism, NADH production was inhibited⁵⁶ using a specific NADH-oxidase (HY-152026, CAS No.: 2762405-17-2, MedChemExpress). Cells were seeded at a density of 105 cells/mL and allowed to adhere overnight. A 5 mM stock solution was prepared in DMSO, according to the manufacturer's instructions. Cells were then incubated with varying concentrations of the inhibitor (0.1, 0.5, 1, 2, 5, and 10 μ M) for 24 h prior to the MTS assay (described in Section [MTS assay](#) section). To account for potential toxic effects, a control group was treated with an equivalent concentration of DMSO, corresponding to equivalent DMSO doses used in the inhibited groups.

Histological processing

Skin samples were removed from the culture insert by carefully cutting the membrane using a scalpel, introduced into a histology cassette, and fixed in 10% neutral buffered tissue formalin (Cat# HT501128-4L, Sigma Aldrich) for 2 hours. The samples were then dehydrated in an automated tissue processor (KD-TS1A, Kedeer Histology Co., LTD) for paraffin inclusion. Samples were finally embedded in paraffin blocks, manually cut into 5 μ m slices using a microtome (MR 3000, Histo-Line Laboratories), and transferred to microscope slides.

Then, tissue slices were deparaffinized using X-free Solvent for histology xylene substitute (Cat# 06-1305Q, Bio-Optica Milano S.p.A.) and successive dilutions of ethanol (Cat# 1009835000, Supelco) to rehydrate the samples. Hematoxylin & Eosin (H&E, Cat# 05-06013/L and Cat# 05-10009/L, Bio-optical, Labolan) staining was performed following the manufacturer's protocol and fixed using mounting medium (Cat# 03989-100ML, Eukitt, Sigma-Aldrich).

MTS assay

To perform the MTS assay (CellTiter 96 aqueous non-radioactive cell proliferation assay, Cat# G3582, Promega, USA), a solution mixing the specific culture medium of the cell type and 10% of MTS reagent was prepared in the dark. Then, this solution was added to the cells and incubated at 37°C for 4 hours. After incubation, the reduced supernatant was collected and transferred to a 96-well plate (Cat# CLS353075, ThermoFisher Scientific) for measurement. The colorimetric reading was performed by measurement of absorbance in a Synergy™ HTX Multi-Mode Microplate Reader, mixing the samples with an orbital spin of 200 rpm and using a wavelength of 490 nm.

Cell confluency and MTS electrochemical signal correlation

To assess the correlation between cell confluency and the measured electrochemical and colorimetric MTS signal, cells were seeded in a 24-well culture plate (Cat# 142475, ThermoFisher Scientific, USA) at increasing confluences from 10% to 100%, based on the cell culture numbers stipulated by the manufacturer. For these specific plates, 100% of confluency corresponds to 240,000 cells, and the remaining cell densities were calculated accordingly, decreasing in 10% increments. The next day, the culture media were removed from the wells and washed with PBS. After that, the MTS assay and electrochemical sensing procedure were performed as explained before (subsection [MTS assay](#) section and subsection [electrochemical measurements](#) section, respectively).

Cell viability assessment after chemical exposure

For the irritation and corrosion assessment in the RHE models, six reference chemicals, selected based on OECD Test Guideline 439: "Performance Standards for the Assessment of Proposed Similar or Modified *in Vitro* Reconstructed Human Epidermis (RHE) Test Methods for Skin Irritation Testing",^{57,58} were employed: isopropanol (Cat# 190764, Sigma Aldrich), 1-decanol (Cat# 150584-500G, Sigma Aldrich), and 1-bromo-4-chlorobutane (Cat# B60800-25G, Sigma Aldrich). These substances were applied topically in their original form (100% concentration) to simulate direct exposure conditions. Additionally, a 5% aqueous potassium hydroxide (KOH) solution (Cat# 221473-25G, Sigma Aldrich) was used as a severe chemical stressor. Sterile PBS was employed as the negative control, while a 5% aqueous solution of sodium dodecyl sulfate (SDS) (Cat# 436143-25G, Sigma Aldrich) served as the validated positive control for the procedure. All liquid solutions were prepared weight/volume (w/v) in deionized water ([Table S2](#)).

Six chemicals extracted from the OECD "Performance Standards for the Assessment of Proposed Similar or Modified *in Vitro* Reconstructed Human Epidermis (RHE) Test Methods for Skin Irritation Testing" as Described in TG 439,^{57,58} were utilized for this

experiment: isopropanol (Cat# 190764, Sigma Aldrich), 5% aqueous potassium hydroxide (KOH) (Cat# 221473-25G, Sigma Aldrich), 1-decanol (Cat# 150584-500G, Sigma Aldrich) and 1-bromo-4-chlorobutane (Cat# B60800-25G, Sigma Aldrich). PBS was used as a negative control, and 5% aqueous solution of sodium dodecyl sulfate (SDS Cat# 436143-25G, Sigma Aldrich) as a positive control for the procedure.

To study the effect of these chemicals on the cells, both hDFs and immortalized N/TERT-1 were seeded at confluency in 96-well plates and incubated for 24 hours at 37°C and 5% CO₂.

On the experiment day, cell culture media were removed from all the wells, and 50 µL of each of the chemicals was applied to the cells for 60 minutes. After the chemical exposition, samples were washed three times with PBS for 5 minutes each, and then a recovery incubation of 42 hours was carried out in the corresponding cell culture medium.

Afterwards, the MTS assay was carried out, including a sample consisting of culture medium and 10% MTS in order to subtract colorimetric and electrochemical background, as defined in the test guidelines.

Cellular viability was measured according to the OECD Guide stated above. The results obtained for the negative control (PBS) were considered as 100% of cell viability, and the rest of the percentages were extrapolated from this value.

Skin irritation assessment after chemical exposure

To assess the skin irritation potential of chemicals in a physiologically relevant *in vitro* model that closely mimics human skin, RHE models were employed. The RHE models were developed using N/TERT-1 keratinocytes as detailed in Section [reconstructed human epidermis cultures](#) and were matured until complete differentiation and cornification (10 days at air-liquid interface) to ensure their suitability for irritation testing. This methodology aligns with the principles established in OECD Test Guidelines for *in vitro* skin irritation (TG 439).⁵⁷

A panel of six test chemicals (isopropanol, 5% aqueous potassium hydroxide (KOH), 1-decanol, 1-bromo-4-chlorobutane, hydrochloric acid (HCl) (Cat# 258148-25ML, Sigma Aldrich) and 10% sulfuric acid (Cat# 1603151000, Supelco), along with PBS (negative control) and 5% aqueous SDS (positive control), was utilized for RHE exposure.

On the day of the experiment, RHE inserts were carefully transferred, and the culture medium was removed from the wells. A volume of 50 µL of each test chemical was applied directly to the apical surface of the RHE model as a standardized exposure period of 60 minutes. Following chemical exposure, the RHE inserts were thoroughly washed three times with PBS for 5 minutes each to remove residual chemicals. A recovery incubation of 42 hours was then performed adding fresh EpiLife medium to allow for potential cellular recovery.

Subsequently, the viability of the RHE models was quantified using the MTS assay (as described in Section [MTS assay](#) section), employing both the electrochemical sensor and conventional colorimetric measurement. Cellular viability for RHE models was measured following the specified OECD Test Guidelines.⁵⁷ The results obtained for the negative control (PBS) were considered 100% viability, and the rest of the percentages were extrapolated relative to this value.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analysis was performed using GraphPad Prism software v.8.0.2 (GraphPad, Dotmatics, USA). All averages and deviations were subjected to 1-way and 2-way ANOVA tests with a significance value of 0.05 to determine the statistical significance of the results. Errors in the readings were calculated using the standard deviations of each measurement. All experiments were performed with sample replications $n = 7$ and RHE samples with $n = 4$. In each figure, the employed test is described in the figure legends. Statistical significance was defined as $p < 0.05$ (*), $p < 0.01$ (**) or $p < 0.001$ (***).